

IP-SAKTI SAHAYAK

FEASIBILITY • VIABILITY • SCALABILITY • IMPACT

SIH TEAM RESEARCH & PRESENTATION GUIDE

THE ONE-LINE IDEA

A trusted, source-cited guidance layer that helps Ayurveda innovators understand which IP, regulatory, biodiversity and traditional-knowledge pathway is relevant—before they act.

IMPORTANT POSITIONING

IP-SAKTI does not replace lawyers, regulators, patent offices, TKDL, IP Saarthi or WIPO PATENTSCOPE. It helps users navigate these systems clearly and safely.

1. WHAT IP-SAKTI IS ACTUALLY SOLVING

The problem is not simply that people cannot find information. An Ayurveda innovation can touch several rule systems at the same time.

- IP: patents, trademarks, geographical indications and traditional-knowledge exclusions.
- Product regulation: medicine/food/product classification and applicable compliance pathways.
- Traditional knowledge: checking whether knowledge is already documented or known.
- Biodiversity / ABS: identifying when biological-resource rules may become relevant.
- Jurisdiction: separating India-specific guidance from international patent or market questions.

CORE DESIGN PRINCIPLE

CLASSIFY FIRST → RETRIEVE AUTHORITATIVE SOURCES → EXPLAIN IN PLAIN LANGUAGE → CITE
→ ABSTAIN WHEN EVIDENCE IS WEAK

2. TARGET USERS & PAIN POINTS

USER	NEED	PAIN POINT
Ayurveda practitioners	Quick, understandable guidance.	Legal language; many portals; limited time.
Researchers / students	Reliable source material for innovation and prior-art exploration.	Scattered documents; difficult cross-linking.
AYUSH / herbal startups	Early IP, product and launch checks.	Classification and compliance uncertainty.
MSMEs / producer groups	Practical guidance before branding or launch.	Limited specialist support.
Farmer / community groups	Understand collective value and protection routes.	Individual vs community protection can be unclear.

3. FOUR HIGH-VALUE USE CASES

01 MSME BRAND CHECK

Before investing in packaging/marketing, an MSME asks what IP route is relevant and which official source to consult.

WHY IT MATTERS: Better-informed branding decisions before money is committed.

02 COMMUNITY / REGIONAL PRODUCT

A farmer cooperative asks whether a distinctive regional product may need community-level protection such as a GI route.

WHY IT MATTERS: Makes collective origin and producer interests visible.

03 STARTUP LAUNCH CHECK

A startup asks food/drug/product and advertising questions before launch.

WHY IT MATTERS: Flags the need for classification and official regulatory verification.

04 RURAL PRACTITIONER

A practitioner asks in simple language and receives a concise explanation with the relevant official source.

WHY IT MATTERS: Reduces the burden of navigating multiple dense portals.

4. TECHNICAL FEASIBILITY

The proposed MVP is feasible as a student-built prototype because it uses established web, backend and retrieval components. The main challenge is integration quality, not technology availability.

LAYER	TOOLS	MVP ROLE
Frontend	React 19 + Vite + JavaScript + Vanilla CSS	Guided chat and result screens.
Backend	FastAPI + Uvicorn + Pydantic + CORS	Validated JSON APIs such as /health and /api/chat.
Knowledge base	PyMuPDF + embeddings + ChromaDB	Extract, chunk and retrieve relevant source passages.
Answering	Gemini Developer API + retrieval pipeline	Generate grounded responses from retrieved evidence.
Classification	Rules engine	Ask focused formulation/product questions before routing.
Verification	Citation + confidence + abstention	Reduce unsupported or overconfident answers.

FEASIBILITY ARGUMENT

Start small: a controlled corpus + a few high-value workflows + citations. Expand only after retrieval and answer quality are tested.

5. ECONOMIC VIABILITY

The prototype can be built economically because the initial architecture can run with a small curated corpus and existing development tools. Costs rise mainly with hosted infrastructure, API calls, usage and corpus size.

- Prototype: local development and a small source corpus keep infrastructure requirements modest.
- Pilot: introduce hosted storage, authentication, logging and controlled API usage as needed.
- Scale: optimize retrieval, caching, model usage and document processing.
- Validation: measure real user tasks, retrieval quality, usefulness and failure cases before claiming economic impact.

DO NOT CLAIM

Do not invent percentage savings, guaranteed ROI, adoption numbers or accuracy percentages. Treat these as outcomes to measure in a pilot.

6. SCALABILITY

- **MORE DOCUMENTS** — Add laws, rules, official guidance and verified documents through a versioned ingestion pipeline.
- **MORE LANGUAGES** — Add translation and voice layers after the core workflow is reliable.
- **MORE USERS** — Add authentication, logging, caching, rate limits and stronger infrastructure.
- **MORE JURISDICTIONS** — Keep India and international pathways separated.
- **MORE COMPLEX QUESTIONS** — Move toward multi-hop reasoning and a knowledge graph in later phases.

SCALABILITY TEST

The system should grow without losing traceability: important answers should still point back to the source used.

7. ADOPTION BARRIERS & MITIGATION

BARRIER	RISK	MITIGATION
Legal trust	AI may be distrusted for legal/regulatory questions.	Official citations + confidence + disclaimer + human review.
Source freshness	Rules and guidance can change.	Source register + version/date + review workflow.
Language	Legal material can be difficult to understand.	Plain-language summaries; multilingual rollout.
Digital literacy	Complex portals can be hard to navigate.	Guided questions and minimal steps.
Connectivity	Remote users may have unreliable internet.	Lightweight UI and future assisted workflows.
Overconfidence	AI can sound certain without evidence.	Abstention rules and escalation.

8. EXISTING SYSTEMS VS IP-SAKTI

SYSTEM	WHAT IT DOES	MAIN SCOPE	IP-SAKTI ROLE
IP Saarthi	AI assistant for broad IP queries.	Broad IP scope.	AYUSH-specific guided classification + cross-domain navigation.

TKDL	Defensive documentation of Indian traditional medicinal knowledge.	Traditional knowledge; controlled access.	Guides users to understand when traditional-knowledge checks matter; does not replace TKDL.
WIPO PATENTSCOPE	Search platform for patent collections and PCT material.	Patent searching.	Guides non-specialists toward relevant searches and explains context.

IMPORTANT

IP-SAKTI should be presented as a complementary guidance layer—not as a replacement for these official systems.

9. EXPECTED IMPACT

DIMENSION	MECHANISM	EXPECTED VALUE
SOCIAL	Easier access to understandable IP and regulatory information.	Better-informed next-step decisions.
ECONOMIC	Better preparation before branding, product development and launch.	Potentially less confusion and rework; actual savings must be measured.
CULTURAL	Greater awareness of traditional knowledge and collective heritage.	Supports responsible handling and protection.
ENVIRONMENTAL	Can surface biodiversity / ABS considerations.	Encourages responsible compliance.
INSTITUTIONAL	Reusable, source-backed knowledge-navigation foundation.	Can connect users to official authorities and documents.

10. WHAT TO PUT ON THE SIH PPT

- **PROBLEM** — Ayurveda innovation crosses IP + regulation + traditional knowledge + biodiversity pathways.
- **USERS** — Practitioners, researchers, startups, MSMEs and producer/community groups.
- **SOLUTION** — Guided classification → official retrieval → plain-language explanation → citations → safe abstention.
- **FEASIBILITY** — Existing React/FastAPI/retrieval stack + phased MVP.
- **VIABILITY** — Small corpus and controlled workflows first; validate before scaling.
- **SCALABILITY** — Documents, languages, users and jurisdictions through modular phases.
- **TRUST** — Source links, confidence, source dates, disclaimers and human review.

- **COMPETITION** — Complement IP Saarthi, TKDL and PATENTSCOPE rather than replacing them.

BEST PITCH LINE

“We are not building another legal portal. We are building the guided layer that helps an Ayurveda innovator know what to check, where to check it, and when not to trust an automated answer.”

11. SOURCE BANK

Ministry of AYUSH — Official ministry portal — <https://ayush.gov.in/>

IP Saarthi / IP India — Official IP India AI-powered IP assistant — <https://ipindia.gov.in/ipsaarthi/>

TKDL — Official TKDL information and access context —
<https://www.tkdl.res.in/tkdl/langdefault/common/Abouttkdl.asp>

WIPO PATENTSCOPE — Official WIPO patent search platform —
<https://www.wipo.int/en/web/patentscope>

IP India — Official IP rights, acts, rules and services — <https://ipindia.gov.in/>

TEAM NOTE: Use official sources for factual claims. If a number, accuracy percentage, adoption figure or cost saving has not been measured, do not present it as a result.

12. ONE-PAGE TEAM CHEAT SHEET

FEASIBLE

Established web, backend and retrieval components. Build a small controlled version first.

VIABLE

Useful workflows exist; validate real economic value through pilot tasks.

SCALABLE

Modular ingestion, languages, users and later reasoning layers provide a growth path.

TRUSTWORTHY BY DESIGN

Citations + source dates + confidence + abstention + human review.

DIFFERENTIATOR

AYUSH-specific classification-first guidance across IP, regulation, traditional knowledge and biodiversity.

IMPACT

Better information access, better preparation, stronger traditional-knowledge awareness and responsible biodiversity decisions.